

Panel Preparation

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Panel Saw Operation

1. The panel saw can cut in two directions: horizontally or vertically. The operation is slightly different for these two kinds of cuts, so it is important to keep in mind which one you are performing.
2. In all cases, let the saw do the cutting; do not force it through the material, or the material through the saw. Check each piece of paneling for staples: running one staple through the saw will chip a carbide tooth, and soon the blade will be worthless.
3. Before loading the first panel onto the saw carriage, TURN ON the dust collector. The switch for this is behind the saw frame on the left, where it will be inaccessible once you load a panel onto the saw.
4. For horizontal cuts, pull the two black knobs on the saw turntable toward you, and rotate the table so that the arrow labeled FEED points to the left. When you release the locking knobs, they should snap into place, and the table should not rotate any further. Then, set the height of the cut, either with a jig or by using the (hidden) scale at the left of the saw table. Lock down the carriage by tightening the black handles on each round guide rail.
5. For vertical cuts, pull the two black knobs on the saw turntable and rotate the table so that the arrow labeled FEED points DOWN. Loosen the locking handles on the guide rails, so that the saw table is free to move vertically, and raise it all the way up.

Roof Panels

1. The material for the exterior is $\frac{3}{4}$ " CDX plywood. Neither side is very pretty, but one is usually worse. That side should face you when sawing roof panels.
2. You will only need to make vertical cuts for this process. Set a block to act as a saw stop for $77\frac{3}{4}$ "
3. Cut 4 panels for each home, placing them on an orange panel cart. Generally, put a maximum of 8 panels on a cart.
4. Secure the panels to the cart with a strap before moving the cart.

Exterior Panels

1. The material for the exterior is T1-11 siding. It is fairly flexible, and requires some care to make sure it is kept flat as it is being sawed.
2. You will need one orange panel cart for each home's panels, and one extra one, to hold panels that will get window cutouts.
3. All panels are to be trimmed down from their initial 96" length to 93½". There is a panel-stop labeled Exterior Panel 93 ½, with two feet sticking out the bottom; these are labeled 1 and 5. Find this panel-stop, and insert it with the feet between the feed rollers on the right carriage of the panel saw, so that the blue tape on the panel-stop matches the blue tape on the panel saw carriage. (The numbers on the bottom of the panel-stop should also match those on the panel saw.)
4. Make sure the saw is set for vertical cuts, and that the saw table is all the way up. Place a panel (good side away from you) on the panel saw rollers on the left carriage, and slide the panel into the saw until it touches the panel-stop.
5. Turn on the saw (green button), place a foot on the panel OUTSIDE the left guide rail, flattening the panel, and lower the saw table to cut the panel. When you reach the bottom, turn off the saw (red button), and wait for the saw blade to stop before raising the table back to the top of its range.
6. Remove the panel, and place it on one of the panel carts.
7. Each home needs 8 solid panels and 2 panels with windows cut out of them. When you have cut 8 panels and loaded them onto a cart, strap the panels to the cart before moving it. Then cut two more panels and put them on the window cart.
8. Repeat 4-7 for each home. All of the cutoff pieces are trash.
9. There is a separate jig for window cutouts. For this process, you will need a router, a drill with a spade bit in it, and a broom and dustpan.
10. Move the cart you have designated for window panels up fairly close to the jig, and bring one of the home-panel carts next to it.

11. Place a panel on the jig, right side up, and with the tongue against the stop. (The tongue is the edge that is thinner than the main panel, and would accept another panel overlaying that thin part.) Make sure the panel slides under the blocks at the upper right corner, and seats both along the long tongue edge and at the top of the panel. Clamp the panel using the fixed clamps, and also use a clamp on the edge opposite the tongue, where it says CLAMP on the jig.
12. Bore an access hole where it is indicated by two arrows on the jig, and rout out the window.
13. Unclamp the panel and place it on the home-panel cart, with the tongue up.
14. Repeat steps 11-13. You should end up with a total of 8 solid panels and 2 window panels on each home-panel cart. Re-secure these panels to their cart.
15. Repeat the window cutout process for each home-panel cart.
16. When you are finished, sweep up as much of the sawdust as you can get to easily, and then use the shop vacuum to collect any remaining sawdust in corners of the jig.

Interior Panels

1. The material for the interior is 15/32" plywood. The good side goes against the carriage when sawing.
2. You will need one orange panel cart for each home's panels, and one extra one, to hold panels temporarily.
3. Set the saw table so that the arrow labeled FEED points to the left.
4. Horizontal cuts. There are jigs to set the saw table height for each panel size. To use one, set the end marked DOWN on the yellow tabs at the bottom of the saw frame between the two guide rails, and then lower the saw table until the bottom edge of the table rests on the ledge of the jig. The words "tuck under bottom of saw carriage" should NOT go under the table, but rest in front of the saw table. Lock the saw table in place, and remove the jig.
5. Cut panels in this order, and stack them on a temporary cart. This list will give you the necessary panels for 2 homes.
 - a. 43 1/2" – 6 panels
 - b. 44" -- 4 panels
 - c. 38 1/4" – 7 panels
6. Reset the saw for vertical cuts. Then, for each home, cut panels according to the following list, and stack them in that order on a cart per home. For most cuts, find the correct panel-stop, and insert it with the feet between the feed rollers on the right carriage of the panel saw, so that the blue tape on the panel-stop matches the blue tape on the panel saw carriage. (The numbers on the bottom of the panel-stop should also match those on the panel saw.)
7. For the E panels, set a block on the right side of the carriage for a 46 1/4" cut. Clamp this block to the frame, and cut one 38 1/4" panel to this width. Cut the remainder to the same width, mark them, and set the two panels aside.

8. Cut panels and mark them IN PENCIL according to this list, using the panel-stops for 86 ½", 88", and 92" panels:

Width	Length	Label	Source
48"	88"	D	Uncut panel
38 ¼"	46 ¼"	E	Step 7 – already cut
48"	88"	G	Uncut panel
38 ¼"	88"	H	Temporary cart (thinnest panels)
44"	86 ½"	C	Temporary cart (widest panels)
48"	92"	A	Uncut panel
38 ¼"	92"	B	Temporary cart (thinnest panels)
44"	86 ½"	C	Temporary cart (widest panels)
48"	92"	A	Uncut panel
38 ¼"	92"	B	Temporary cart (thinnest panels)
43 ½"	44"	R2	Temporary cart (middle width panels)
43 ½"	44"	R2	Remainder from previous R2
43 ½"	92"	R1	Temporary cart (middle width panels)
43 ½"	92"	R1	Temporary cart (middle width panels)

9. You should end up with the two R1 panels at the top of the pile. Strap the panels to the cart before moving the cart, and repeat for each home.